

Windows vs Linux

Comparative Study

1. Comparative Table

Criteria	Windows	Linux
Operating system type	Proprietary operating system	Open-source operating system
Developer	Microsoft	Open-source community
License	Paid license	Free (GPL license)
Source code	Closed source	Open source
Ease of use	Very user-friendly	Depends on the distribution
Graphical interface	Single interface (Windows Desktop)	Multiple desktop environments (GNOME, KDE, XFCE...)
Customization	Limited	Very high
Security	Good, but often targeted by malware	Very secure, few viruses
Software installation	Executable files (.exe, .msi)	Package managers (APT, DNF, etc.)
Updates	Windows Update	Centralized via package manager
Hardware compatibility	Excellent	Very good
Performance	Requires more resources	Lightweight and efficient
Use in companies	Very common	Very common on servers
Server usage	Windows Server	Widely used for servers
Examples	Windows 10, Windows 11	Ubuntu, Debian, Fedora

2. Advantages and Disadvantages

Windows

Advantages

- Easy to use
- Wide software compatibility
- Common in professional environments

Disadvantages

- Paid license
- Less customizable
- More exposed to viruses

Linux

Advantages

- Free and open source
- Very secure and stable
- High performance
- Ideal for learning operating systems

Disadvantages

- More complex for beginners
- Some proprietary software unavailable
- Hardware support can vary

3. Conclusion

Windows is designed for **general users**, while Linux is designed for **system control and flexibility**. Both operating systems are important to know for IT students.

4. Multiple Choice Questions (MCQ)

Q1. Which operating system is open source?

- a) Windows
- b) macOS
- c) Linux
- d) Android

Q2. Which company develops Windows?

- a) Apple
- b) Google
- c) Microsoft
- d) IBM

Q3. Which operating system is mostly used on servers?

- a) Windows Home

- b) Linux
- c) macOS
- d) Android

Q4. What is a package manager?

- a) A virus protection software
- b) A hardware driver
- c) A system update tool
- d) A software installation system

Q5. Which OS usually requires a paid license?

- a) Linux
- b) Ubuntu
- c) Debian
- d) Windows

5. Open Questions

1. Explain the difference between **open source** and **proprietary software**.
 2. Give **two advantages** of Linux for servers.
 3. Why is Windows widely used in companies?
 4. Name **two Linux distributions** and explain their use.
 5. Which operating system would you choose for an old computer? Explain why.
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